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Validation of an Electronic Health Record-Based Machine Learning Model Compared to Clinical Risk Scores for Gastrointestinal Bleeding

Dennis L. Shung[1,9,10,*], **Colleen E. Chan**[2], **Kisung You**[3], **Shinpei Nakamura**[2], **Theo Saarinen**[4], **Neil S. Zheng**[1], **Michael Simonov**[1], **Darrick K. Li**[1], **Cynthia Tsay**[5], **Yuki Kawamura**[6], **Matthew Shen**[4], **Allen Hsiao**[1], **Jasjeet Sekhon**[2,7,†], **Loren Laine**[1,8,†]

[1]Section of Digestive Diseases, Department of Medicine, Yale School of Medicine, New Haven CT USA

[2]Department of Statistics and Data Science, Yale University, New Haven CT USA

[3]Department of Mathematics, City University of New York Baruch College, New York NY USA

[4]Department of Statistics, University of Berkeley, Berkeley CA USA

[5]Department of Gastroenterology, Johns Hopkins School of Medicine, Baltimore MD USA

[6]University of Cambridge School of Clinical Medicine, Cambridge UK

[7]Department of Political Science, Yale University, New Haven CT USA

[8]West Haven Veterans Affairs Medical Center, West Haven CT USA

[9]Clinical and Translational Research Accelerator, Department of Medicine, Yale School of Medicine, New Haven CT USA

[10]Department of Biomedical Informatics and Data Science, Department of Medicine, Yale School of Medicine, New Haven CT USA

Abstract

Background & Aims—Guidelines recommend use of risk stratification scores for patients presenting with gastrointestinal bleeding (GIB) to identify very-low-risk patients eligible for discharge from emergency departments. Machine learning models may outperform existing scores and can be integrated within the electronic health record (EHR) to provide real-time risk

[*]**Correspondence:** Dennis L. Shung MD MHS PhD, PO Box 208019, New Haven CT 06520-8019, dennis.shung@yale.edu.
[†]Denotes co-senior authorship

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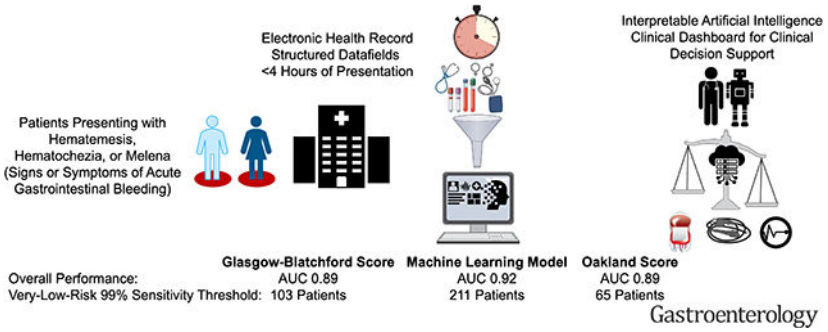
assessment without manual data entry. We present the first EHR-based machine learning model for GIB.

Methods—The training cohort comprised 2,546 patients and internal validation of 850 patients presenting with overt GIB (hematemesis, melena, hematochezia) to emergency departments of 2 hospitals from 2014–2019. External validation was performed on 926 patients presenting to a different hospital with the same EHR from 2014–2019. The primary outcome was a composite of red-blood-cell transfusion, hemostatic intervention (endoscopic, interventional radiologic, or surgical), and 30-day all-cause mortality. We used structured data fields in the EHR available within 4 hours of presentation and compared performance of machine learning models to current guideline-recommended risk scores, Glasgow-Blatchford Score (GBS) and Oakland Score. Primary analysis was area under the receiver-operating-characteristic curve (AUC). Secondary analysis was specificity at 99% sensitivity to assess proportion of patients correctly identified as very-low-risk.

Results—The machine learning model outperformed the GBS (AUC=0.92 vs. 0.89;p<0.001) and Oakland score (AUC=0.92 vs. 0.89;p<0.001). At the very-low-risk threshold of 99% sensitivity, the machine learning model identified more very-low-risk patients: 37.9% vs. 18.5% for GBS and 11.7% for Oakland score (p<0.001 for both comparisons).

Conclusions—An EHR-based machine learning model performs better than currently recommended clinical risk scores and identifies more very-low-risk patients eligible for discharge from the emergency department.

Graphical Abstract



Keywords

electronic health record; machine learning; gastrointestinal hemorrhage

Introduction

Acute gastrointestinal bleeding (GIB) is the most common gastrointestinal diagnosis requiring hospitalization in the U.S., accounting for 2.2 million hospital days and inpatient charges of $19.2 billion dollars.[1] Traditionally acute GIB has been classified by the anatomical location of the bleeding, either upper or lower GIB.[2, 3] Guidelines recommend risk stratification using clinical risk scores for both upper and lower GIB to identify very-low-risk patients, defined as patients who do not require red-blood-cell transfusion

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or intervention to stop bleeding, and who do not die.[4–6] Once accurately identified as very-low-risk, such patients can then be considered for discharge from the emergency department with outpatient management, thereby reducing costs without risk of harm to the patient.[7] Potential cost savings with use of this management strategy for very-low-risk patients presenting with upper GIB have been estimated at more than $3.4 billion over 5 years in the U.S.[8] However, existing clinical risk scores for upper and lower GIB are used uncommonly in clinical practice.[9] Barriers to use include lack of awareness of guidelines, cumbersome data entry, an uncertain distinction between upper and lower sources of GIB at presentation, clinician information overload, as well as relatively poor real-world performance of many risk scores.[10]

The location of the bleeding source is not always clear at presentation: a patient presenting with melena may have a bleeding lesion in the small intestine or proximal colon, while hematochezia can be the presenting symptom for up to ~15% of patients with upper GIB.[11] Therefore, risk assessment tools that are designed for patients with symptoms of GIB rather than site of bleeding would be more clinically useful and practical for assessment of risk in the emergency department. The Glasgow-Blatchford Score (GBS) has been validated on multiple patient cohorts with upper GIB[12, 13] and is able to identify very-low-risk patients with high sensitivity (low false negative rate)[14, 15] allowing discharge of these patients from the emergency department without hospitalization in a prospective clinical trial.[7] The Oakland score is the best performing clinical risk score recently developed for lower GIB [16, 17] and has shown good performance for identifying very-low-risk patients when externally validated in a U.S. electronic health record (EHR) database.[18]

Machine learning, a field of study that gives computers the ability to learn adaptive and accurate models from data, can be used to construct models that perform better than clinical risk scores for GIB.[19, 20] The EHR stores a large quantity of clinical data that can be used to automatically calculate risk assessments and has been used to train models for risk prediction across multiple clinical scenarios.[21, 22] Ideally, such a model could be applied to all patients with GIB, preventing the need for immediate differentiation between upper and lower GIB. We previously showed a machine learning model performed better than standard risk stratification tools in identifying very-low-risk patients with upper GIB, but this analysis used only 24 variables obtained by manual data collection.[19]

We propose a symptom-based machine learning risk model derived from EHR data that bases initial assessment on presenting symptoms rather than presumed anatomical location. This score is designed to be deployed automatically through the EHR by extracting data to automatically calculate scores that can then be made available to providers in real-time, soon enough after presentation to be of clinical utility in making important decisions regarding care. We compare the best performing machine learning model to a deep learning model and two guideline-recommended clinical risk scores, the GBS and the Oakland Score. Due to a concern for systemic bias that may affect access to endoscopic evaluation in patients with different race/ethnicity[23], we performed equity evaluation across biological sex, race and ethnicity. To address the issues of interpretability and trust necessary for successful implementation, we designed an artificial intelligence clinical decision support system (AI-CDSS) for the machine learning model with an interpretable interactive dashboard.

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Methods

Data Source

A cohort of 2,546 patients presenting with overt GIB, defined as hematemesis (either bright red blood or coffee ground emesis), melena, or hematochezia, from July 2014 to January 2019 to the emergency departments of the two campuses of Yale New Haven Hospital in New Haven, CT was used to train the machine learning model, with an internal validation holdout set of 850 patients used to tune the machine learning model. A separate cohort of 926 patients presenting with overt GIB to the emergency department of a separate tertiary care urban hospital with the same EHR (Bridgeport Hospital, Bridgeport, CT) from July 2014 to January 2019 was used for external validation. We note that the separate hospital shares no gastroenterology faculty with the first two campuses, and represents a geographically distinct patient population in a separate urban center. The cohort was identified through a process of EHR phenotyping, complete case analysis, and manual expert review of the note text at presentation. We created a sensitive data mart, a phenotyping method developed to handle the velocity, variety, and volume of electronic health record data that we had applied in a previous work to identify patients with acute GIB.[24, 25] We identified a cohort of patients from 2014 to 2019 identified with mention of the concept of GIB using EHR phenotyping (N=23,319). EHR phenotyping for overt GIB using ontology mapping with SNOMED-Clinical Terms for GIB, a Boolean decision rule integrating GIB-specific triage terms (structured datafields entered by the emergency department triage nurse) and GIB-specific review-of-systems fields identified a cohort of patients who had acute GIB at these hospitals from 2014 to 2019. Patients with complete vital sign data measured within 4 hours (N= 22,834) were further subset to patients with laboratory values measured within 4 hours (N=11,009). All charts were then manually reviewed by two board-certified gastroenterologists for patients who had actual overt signs or symptoms of GIB in the emergency room provider note, as opposed to developing signs or symptoms of overt GIB during the hospital stay (N=4,322). The final cohorts included 2,546 for training, 850 for internal validation, and 926 patients for external validation. (Table 1)

Study Design

Outcome Variable—The primary outcome was a composite of red-blood-cell transfusion, hemostatic intervention (endoscopic, interventional radiologic, or surgical), and 30-day all-cause mortality.

Input Variables—The final dataset used for modeling included 2 demographic variables, 8 initial vital signs-related variables, 3 nursing assessment variables, 20 laboratory test results initially measured in the first 4 hours, 48 mapped medical history diagnoses, and 25 medication classes (Table 2). This set of input variables were selected from a total of 5,281 unique structured data fields in the EHR (Epic, Verona, WI) that were available within 4 hours of presentation to the emergency department. We selected the specific laboratory values and flowsheet values due to the plausible links to known predictors from previous studies for hospital-based intervention in patients with either UGIB or LGIB,[12, 17, 20, 26–30] and included elements that were routinely available as part of order panels in the emergency department (e.g. liver tests including total bilirubin, complete blood count

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with differential, complete metabolic panel including creatinine and blood urea nitrogen, prothrombin time and international normalized ratio). For medication classes and prior diagnoses, we employed supervised variable selection to identify the input variables that are most predictive of the composite outcome.

Our initial variable selection employed LASSO regression, which identifies linear predictors for GI bleeding risk by penalizing less significant variables. This method led to the inclusion of variables based on their linear predictive power. When later inputting these variables to an XGBoost model, which captures complex nonlinear interactions, the direct contribution of each variable is more complex and does not independently contribute to the predictability of the model. We decided to employ this method after comparing a variety of other supervised and unsupervised methods to identify predictive variables as detailed in Appendix A1–A2 to identify the final input variable set.

Model Background and Comparisons—We applied five deep learning and machine learning estimators: gradient boosted trees, regularized regression with the elastic net penalty, the random forest algorithm, 2-layer neural networks, and 5-layer neural networks. These models were run using the glmnet[31], xgboost[32], Rforestry[33], and tensorflow packages and tuned using the caret package[34] in R (R Foundation for Statistical Computing, Vienna, Austria). The gradient boosted tree model assembles a collection of decision trees (tree ensemble model) by adding trees that minimize prediction error measured by a gradient at each training step. Specifically, the model used also has hyperparameters for regularization, shrinkage, and subsampling to prevent overfitting and reduce computation time. The random forest model combines the predictions of an ensemble of decision tree learners, each trained on a bootstrap sample of the original training data. The random forest model selected makes use of honesty – the practice of using disjoint subsets of the training data for determining the tree structure for each tree in the forest and producing the predictions from each tree. This practice guards against the forest overfitting the training data and reduces the variance of the predictions when the forest is applied to new data. For the deep learning neural network model, we trained two feedforward neural networks: one with 2 layers and another with 5 layers with ReLU activation functions attached to a binary classifier to predict outcome. Prior to this choice we varied the number of layers from 2 to 5 without much difference in performance even after tuning over a hyperparameter set (different activation functions, levels of dropout at each layer, number of neurons in each layer, optimization functions, and cost functions). We performed a randomized grid search to find the optimal hyperparameters, which is further detailed in Appendix A3.

The estimators were applied on several data subsets, which varied not only in the types of personal mediation history codes used, such as ICD codes, Phecodes, CCS codes, and custom-developed codes, but also occasionally included medication classes. All data subsets consisted of a set of base variables (Table A1.1), which included demographics, lab tests, nursing assessment variables, and vital signs. This process is described in further detail in Appendix A1–A2.

Clinical risk scores used for comparisons include the Glasgow-Blatchford Score and the Oakland Score (full list of components in Appendix A5). Thresholds used to calculate the

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scores used the original criteria[35] whereas the Oakland score was calculated by removing the rectal exam since the largest study performed in the U.S. validated a version of the modified score.[18] For clinically determined concepts (e.g. liver failure, melena, cardiac disease), manual labeling by a gastroenterologist was used.

Statistical Analysis

Primary analysis of model performance was assessment of area under receiver operating characteristic curve (AUC), with a predefined goal of AUC>0.90. The AUC on internal validation was reported with a randomly split test set, and the AUCs were compared using the nonparametric DeLong test. Secondary analysis was specificity when sensitivity is 99% or higher. Specificity identifies the proportion of patients who will not die or need transfusion or intervention (i.e., very-low-risk patients) who are correctly identified by the risk score as being at very-low-risk; the higher the specificity the more patients who can be discharged from the emergency department with very low risk. Comparison of specificities was performed at the 99% sensitivity threshold for external validation (versus GBS=0, Oakland=8). We also performed a comparison at the matched sensitivity of GBS ≤1, which is the recommended very low risk threshold for guidelines for upper GIB. Specificities were compared using McNemar's test. Chi-square and t-tests were used to compare the variables and characteristics of training and test patient cohorts. Calibration was measured by Brier score, a measure of usefulness for predicted probability by evaluating accuracy of predicted probability compared to observed probability.[36] The calibration scores for the machine learning model, the GBS, and the Oakland score are provided with additional details in Appendix A4. Pre-specified post-hoc analysis by biological sex, race, and ethnicity was performed for equity evaluation.

Interpretability Methods

Predictive modeling with complex models like random forests, gradient boosted trees, or deep learning models are challenging to understand. Understanding how input variables influence predictions can help with interpreting machine learning output. Interpretable machine learning refers to methods to explain the prediction, in this case the risk of patients requiring a hospital-based intervention or dying. The importance of interpretability is particularly acute in time-constrained and high-stakes situations where clinicians need to trust clinical decision support systems. When predicting the risk of GIB, it is important to match a clinical mental model of risk to the risk output from the machine learning model by explaining both average effect of individual variables (using partial dependence plots) on risk and the individualized risk for each patient (using individual conditional expectation plots). We include details about the rationale for and definitions of interpretable machine learning methods in the Appendix A6. We present a customized interface with both types of plots of the top features in our final model in the interactive interface in Appendix A7.

Human-Algorithmic Interaction

AI-CDSS are the latest iteration of clinical decision support systems used in clinical medicine to assist human clinical providers in complex clinical decision-making.[37] To be effective, AI-CDSS must be effectively integrated to realize the potential benefits for clinical care. Principles for successful integration in clinical environments include

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partnering with, not aiming to replace clinicians [38] and providing explanations by justifying recommendations as part of their design.[39] Interpretable machine learning techniques have emerged as a subfield that provides insight into the behavior of the model over different ranges of input variables with software packages that provide visualizations that are suitable for human interpretation.[40, 41] (Appendix A6) We developed an interpretable machine learning dashboard integrating state-of-the-art methods to help guide provider decision-making for GIB risk. (Figure 1)

Results

The training and internal validation sets were from the same centers and randomly split, with similar demographic and laboratory values at presentation. The external validation was from a separate center and had higher mean values for liver tests (ALT, AST, total bilirubin), lower proportion of the population identifying as Caucasian (59% versus 66–70%), and slightly higher mean INR (1.12 versus 1.04–1.07). The comparison of GBS and Oakland Scores and component breakdown of each score in the external validation dataset are shown in Table 3.

External Validation

The AUC for the machine learning model (AUC=0.92, 0.90–0.95) was similar to the deep learning model (AUC=0.92, 0.90–0.94; p=0.28) and was higher than GBS (0.89, 0.87–0.92; p<0.001) and Oakland score (0.89, 0.86–0.92; p<0.001) (Table 4).

At 100% sensitivity, the machine learning model has higher specificity than Oakland=4 (10.4% vs 0.5%, p<0.001). No comparable threshold was available for GBS since GBS=0 had a sensitivity of 99%. (Table 5).

At 99% sensitivity, the machine learning model has higher specificity than Oakland=8 (37.9% vs 11.7%, p<0.001) and higher specificity than GBS=0 (37.9% vs 18.5%, p<0.001). At 99% sensitivity the machine learning model also outperformed the 5-layer deep learning model (37.9% vs 28.2%, p = 0.002).

At 99% sensitivity, the machine learning model correctly identified more very-low-risk patients (N=211) among the 926 patients in this cohort than Oakland=8 (N=65), GBS=0 (N=103), and deep learning model (N=157). (Table 5). (Table 5)

For these cutoffs we present a flow diagram that compares the patients in the external validation identified as true positives, false negatives, true negatives, and false positives for the machine learning model at 99% sensitivity, deep learning model at 99% sensitivity, Glasgow-Blatchford Score=0, and Oakland Score=8. (Figure 2) Based on the 99% sensitivity threshold, predictions were either classified as “very low risk” or “not very low risk”. This threshold ensures the number of true positives–patients requiring a hospital-based intervention correctly classified as “not very low risk”–was at least 99% across all models and scores. The advantage of our selected machine learning is the superior rate of identifying true negatives, or patients who do not require a hospital-based intervention and were correctly classified as very low risk. The machine learning correctly identified 211 of

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the 557 low risk patients (37.9%), compared with 157 patients (28.2%) for the deep learning model, 103 patients (19.6%) for the GBS, and 65 patients (11.7%) for the Oakland score. Table 5 provides a statistical comparison of these specificities to that of the machine learning model using McNemar's test, where all comparisons yielded a p-value < 0.01.

Since the guidelines for UGIB vary in recommending either a 99% sensitivity threshold or a strict point cutoff (GBS≤1), we also tested a matched sensitivity threshold with GBS≤1 (sensitivity 98.9%); the machine learning model outperformed GBS≤1 in specificity (39.1% vs 33.3%, p=0.003). At the matched cutoff for GBS≤1 the machine learning model identified more very-low-risk patients (N=218) versus GBS≤1 (N=186). We note that the model output threshold for 99% sensitivity is the raw probability output of 6.3% on the internal validation set, which was the probability output threshold we applied on the external validation set.

We performed a sensitivity analysis to address the possibility of patients who are unknown to the health system presenting with acute GIB have no past medical history and no medications entered in the EHR. We assumed that the only variables available were laboratory variables, vital sign variables, demographic variables, and nursing assessment variables within the first 4 hours (base variable set, Appendix Table 1.1). The best performing model trained on this subset of variables had similar external validation performance to the best overall model in terms of overall AUC (0.92 vs 0.92;P=0.13) but lower specificity at 99% sensitivity (27% vs 37.9%;P<0.001).

Calibration

The calibrated outputs of the final ML model (Brier Score 0.10) was better than GBS (Brier Score 0.13) and Oakland score (Brier Score 0.13). (Appendix A4)

Disparity Analysis

We examined the performance of the model across biological sex, race, and ethnicity, which were not used in the training of the model and found that performance was similar across categorized biological sex, race, and ethnicity. (Appendix A8)

GIB Symptom Subgroup Analysis

We examined the performance of the model across hematemesis, hematochezia, and melena and found similar performance. (Appendix A8)

Error Analysis

We evaluated the patients at the 99% sensitivity threshold where the machine learning model identified false positives, where it predicted a higher risk for a very low risk patient who did not in fact require a hospital-based intervention. We compared the characteristics of these patients compared to the true negatives, or patients who were identified as very low risk who did not require a hospital-based intervention and found marginal differences that may explain the mis-classification, including lower systolic blood pressure, hemoglobin and hematocrit. The differences are not clinically significant despite statistical differences, which suggest the need for human judgment in conjunction with the machine learning risk prediction for optimal clinical management. For all patients who were mis-classified, most

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were very close to the very low risk probability threshold; we present the comparisons in Appendix 9.

Discussion

To our knowledge, this is the first study to develop and validate an EHR-based machine learning model that provides excellent performance in predicting very-low-risk patients presenting with acute GIB who are eligible for discharge from the emergency department without admission to hospital. This machine learning model demonstrates similar overall performance to the deep learning model and superior overall performance as compared to the two guideline-recommended clinical risk scores, the GBS and the Oakland Score. Surprisingly, we found that among this cohort of patients presenting with overt symptoms of bleeding from all sites throughout the gastrointestinal tract (melena, hematochezia, hematemesis) the overall performance of GBS and Oakland scores were very good and similar, despite the fact that GBS was developed and validated solely for upper GIB and Oakland score for lower GIB. Our results provide support for the recommendations of prior guidelines to use the GBS and Oakland Score for risk stratification of patients with UGIB and LGIB, respectively.

At the high-sensitivity cutoffs designed to avoid falsely labeling patients who will die or require transfusion or intervention as very-low-risk, the machine learning model identifies more patients who are eligible for discharge and outpatient management than the deep learning model and currently recommended clinical risk scores. On the external validation dataset, the machine learning model identifies approximately twice as many patients who can be safely discharged home compared to the GBS and identifies more than three times as many as the Oakland score. The model was developed specifically on data collected and available in the EHR within the first four hours of presentation, which allows the potential integration of the model in real time to provide decision support. The score can be automatically calculated and does not require manual data extraction and input from the provider due to the use of only structured datafields from the EHR. We found no clear differences of machine learning model performance across biological sex, race and ethnicity.

We tested a total of over 6,000 different models spanning 5 machine learning and deep learning estimators, 50 different hyperparameters, and 27 separate preprocessing methods (9 code combinations and 5 different embedding methods for 2 of those combinations) with substantial multidisciplinary (statistical, clinical, and informatics) work and expertise applied at each step of the iterative process. Our pipeline integrates the preprocessing of individual sparse diagnosis codes and medication prescriptions into known groupers and performs supervised selection of the most predictive variables to include in this final model to maximize model performance. This setup also allows for flexibility in customizing the machine learning model for other sites, which may have different patterns of disease.

Finally, we present the model output in a dashboard as part of an advanced AI-CDSS with state-of-the-art interpretable machine learning methods to promote trust and increase usability (available at: https://dshung.shinyapps.io/EHR-GIB/.) This work serves as foundation for an HER-integrated and validated advanced AI-CDSS for acute GIB,

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addressing many of the previously identified barriers of guideline awareness, clinician information overload, manual data entry, and the challenges of quantifying risk for patients with overt signs or symptoms of acute GIB.

With the advent of more high-performing AI-CDSS, it is conceivable that these systems could be used not only by providers in the emergency department but also providers in outpatient clinics. We caution that our model is developed specifically in a cohort of patients who have already presented to the emergency room, thus selecting specifically for patients whose concern has reached a level of urgency that may not capture the larger population of patients who experience signs or symptoms of acute overt GIB but choose to see their primary care physician or do not seek medical attention. In the future, expansion of our model to include patients who present to outpatient clinics may represent a direction that could inform outpatient management of acute overt GIB. A major barrier to to at-home use of such AI-CDSSs is the absence of laboratory testing and hemodynamic measurement outside of healthcare facilities. In the era of home laboratory testing and wearable devices, there is also an opportunity to imagine a future where initial at-home evaluation could become feasible with subsequent pathways (e.g., emergency room visit, outpatient management) based on at-home AI-CDSS output used by patients directly.

Strengths

While many machine learning models have been developed, the gap between development and implementation is hampered by the choice of dataflow, absence of external validation and calibration, and no consideration for where or how the tool will integrate into and enhance provider workflow. For example, two recent studies of EHR-based machine learning models for acute GIB have examined the use of machine learning tools to predict mortality and need for red-blood-cell transfusion for patients in the intensive care unit.[27, 30] These models may have more limited utility given the fact that most patients with GIB aren't admitted to the intensive care unit and many important care decisions (e.g., admission vs. discharge, level of care if admitted, initial resuscitation and transfusion requirements, initial medications for GIB, timing of endoscopy) are made prior to placement in the intensive care unit.

In contrast, our study specifically targets the guideline-recommended risk stratification triage in the emergency department. We provide a comprehensive scheme for developing and validating an EHR-based deep learning and machine learning tools that are designed for integration into the assessment of patients with acute GIB in the emergency department in 4 hours from presentation and can be used to help decide whether very-low-risk patients can be discharged for outpatient management. Since this model is developed using structured datafields available within 4 hours of the patients' presentation, it can automatically extract the necessary data from the patient record and calculate a risk score after the data are processed using the gradient boosted tree model, thus eliminating the cumbersome application of traditional risk scores. Our pipeline integrates the preprocessing of individual sparse diagnosis codes and medication prescriptions into known groupers and performs supervised selection of the most predictive variables to include in this final model to maximize model performance. This setup also allows for flexibility in customizing the

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machine learning model for other sites, which may have different patterns of disease. This study demonstrates robust performance on internal validation and on external validation in a separate hospital using the same EHR system. This study also calibrates the risk predictions for clinical use and demonstrates low risk of propagating biases that may restrict timely access to care with no differences in performance across biological sex, race and ethnicity.

We also pave the way for real-world deployment for provider use with state-of-the-art interpretable machine learning method-driven dashboard AI-CDSS displaying general and individual-level information for the machine learning prediction (Figure 1). This dashboard can be used by providers to improve interpretability of the model's output with the goal of increasing trust and thereby use of this AI-CDSS. This dashboard will have pre-loaded EHR data to allow users to make predictions with our model, view diagnostic metrics of the model internals such as Partial Dependence (PDP), Accumulated Local Expectation (ALE), and Individual Conditional Expectation (ICE) plots, and query the most important features. In the era of learning health systems, this study suggests that machine learning with EHR data has the potential of scalability not only at one center but multiple centers across a health enterprise. The development of our model in the most widely used EHR system in the U.S. may be useful in generalizing this algorithmic approach to other healthcare systems to provide evidence-based risk assessment. Epic is also most the widely used EHR across the globe[42], and is increasingly present in healthcare systems throughout the world, including Singapore, the United Kingdom, Sweden, the Middle East and Canada. The global presence of Epic leads to the possibility of mapping similar structured datafields across different Epic instances, and our approach builds in the tools necessary to map variables and re-train them across different Epic databases. The path to interoperability remains challenging, however, due to customized builds for each health system and different regulatory structures internationally.

Limitations

Complete case analysis was performed, limiting the use of all training examples. Missingness in EHR data, especially laboratory data, is of particular concern since practice patterns may be biased by experience, seniority, and provider perception. In order to allay those concerns in this analysis, only patients with measured values for laboratory values and vital signs were included in the analysis. The cutoff for 4 hours was determined after discussion with ED providers regarding the timeframe for clinical decision-making and evaluation of multiple cutoffs found within the patient cohort. In future studies, new ways of modeling missing data and comparing the performance of various approaches of data imputation, as well as algorithmic approaches to integrate data collected or updated after a set time period (e.g., 4 hours) will be important to increase representation of all patient data. The findings of this study, while promising, require prospective implementation in live ED provider workflow to evaluate real world performance.

Importantly, feasibility and usability studies that assess provider acceptance are critical. Interpretability of the machine learning tool is also an aspect that needs to be studied, particularly due to the role of the tool in assisting clinicians' decision making. The main challenges to implementing such a model into EHRs include hospital leadership

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buy-in, informatics expertise, and data structure familiarity across the different database configurations within Epic—as well as availability of secure cloud computing to calculate machine learning risk predictions. We acknowledge that different data distributions exist across the USA and globally and anticipate using the same data pre-processing pipeline to re-train our model or adapt the model using data from other health systems to improve generalizability. This could be through transfer learning, data augmentation with synthetic data, or re-training the model from the raw data in the different centers using our pipeline.

Conclusions

We present the first externally validated automated EHR-based machine learning risk stratification model for patients presenting with signs or symptoms of acute overt GIB that identifies more very-low-risk patients than existing clinical risk scores. This model is designed for real-world deployment by using available data integrated into an interpretable dashboard that provides clinical decision support to suggest outpatient management for very-low-risk patients and potentially decrease unnecessary hospitalizations, resulting in value for patients, health systems, and payers.

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Data Transparency Statement:

We will make the de-identified data for the clinical risk scores available, but the full electronic health record data will not be available due to concerns for privacy and risk of re-identification.

APPENDIX

A1 Data Subsets

In our analysis, we utilized several distinct data subsets. Each subset incorporated a foundational set of 33 base variables, which is comprised of 2 demographics variables, 11 nursing assessment values, and 20 laboratory test results. These variables are listed in Table A1.1. To explore the potential influences and interactions with various personal medical history and medication classes, we expanded upon the foundational dataset by also incorporating PHE, CCS, ICD, or our custom-mapped personal medical history codes with and without medication classes in Table A1.2. These different subsets are listed in Table A1.3 and described in further detail in Appendix A2.

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Table A1.1:

Input variables that comprise the foundational set of base variables.

Variable class	Variables
Demographics (2)	Age at encounter Sex
Vital Signs (8)	Diastolic blood pressure Systolic blood pressure Pulse Pulse oximetry Mean arterial pressure Respiratory Rate Shock Index Temperature
Nursing Assessment (3)	Glasgow Coma Score Eye Component Glasgow Coma Score Motor Component Glasgow Coma Score Verbal Component
Lab Values (20)	Albumin Alkaline Phosphatase Alanine Aminotransferase Aspartate Aminotransferase Total Bilirubin Blood urea nitrogen Chloride Bicarbonate Creatinine Hematocrit Hemoglobin International normalized ratio Potassium Mean corpuscular hemoglobin Mean corpuscular hemoglobin concentration Mean corpuscular volume Platelet count Red cell distribution width Sodium White blood cell count Blood Urea Nitrogen / Creatinine

Table A1.2:

custom mapped diagnoses and medication classes

Custom-Mapped Diagnoses	Cerebrovascular Accident Transient Ischemic Attack Deep Vein Thrombosis Pulmonary Embolism Atrial Fibrillation Prior Upper Gastrointestinal Bleed Prior Lower Gastrointestinal Bleed Prior Unspecified Gastrointestinal Bleed Peptic Ulcer Disease H. Pylori Coronary Artery Disease Acute Coronary Syndrome Heart Failure Hypertension Type 2 Diabetes Mellitus Chronic Kidney Disease Alcohol Use Disorder Cirrhosis
Custom-Mapped Medications	Aspirin P2Y12 Inhibitors Warfarin Direct Thrombin Inhibitors NSAIDs

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	Proton Pump Inhibitors H2 Inhibitors Direct Oral Anticoagulants

Table A1.3:

Data subsets considered that include medical history as denoted by available International Classification of Disease (ICD-10) codes alone, expert-selected or grouped as Phecodes (PHE) or Clinical Classification System (CCS), and medication classes. We also introduced a LASSO pipeline for feature selection across these classes.

Dataset	# of variables	Included classes
Base	33	See Table A1
Base + Expert-Selected ICD	50	Base variables (33) Expert-Selected ICD codes (17)
Base + ICD	2600	Base variables (33) ICD codes (2567)
Base + ICD + meds	3016	Base variables (33) ICD codes (2567) Medication classes (416)
Base + PHE	1788	Base variables (33) PHE codes (1755)
Base + PHE + meds	2204	Base variables (33) PHE codes (1755) Medication classes (416)
Base + CCS	523	Base variables (33) CCS codes (490)
Base + CCS + meds	939	Base variables (33) CCS codes (490) Medication classes (416)
Base + Expert/LASSO-selected	79	Base variables (33) Curated Medical History codes (17) LASSO-selected CCS/PHE/ICD codes (15) LASSO and expert-selected medication classes (14)

A2. Feature Engineering

The raw structured datafields include 2 demographic variables (sex, age at encounter), 416 medication classes from current medication list tables, 2,567 unique International Classification of Diseases (ICD)-10 codes, 24 laboratory measurements, and 7 vital signs, and the 3 components of Glasgow Coma Score from nursing assessments. For data subsets with a high number of variables, we applied five embedding methods to the appended variables classes. The objective of employing these embedding methods was to reduce the dimensionality of the data while retaining the most important features or structures of the data for improved downstream analyses. These are described in further details in Appendix A2.

Feature Engineering of Input Variables

Feature selection to the final set was a combination of expert-driven selection of relevant laboratory values, vital signs, and provider assessment plus a mapping and consolidation of

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features from the medical history and medication tables. The final dataset includes 20 base laboratory tests, 6 variables from 3 vital signs, 3 variables from nursing assessments, age, gender, 1 variable based on transformations of laboratory test values, 1 variable based on transformations of vital sign values. Nursing assessments were based on the Glasgow Coma Score and coded into a categorical variable using one hot encoding. In order to make use of the medical history, several feature engineering schemes were applied to ICD-10 codes listed in the medical history. Below we detail the different selection methods that were tried but not used for the final model.

ICD-10 Codes

ICD-10 codes for inclusion in the model were determined by their occurrence in the training data. Codes that appeared at least once in a patient within the training data were included. Based on this criterion, 181 ICD-10 codes were selected from the 2,546 patients in the training set.

LASSO-selected/Expert Mapping

ICD-10 codes from the medical history were mapped to a curated set of highly relevant diagnoses selected by gastroenterologists, including stroke, venous thromboembolism, atrial fibrillation/flutter, peptic ulcer disease, *H. pylori*, coronary artery disease, heart failure, hypertension, type 2 diabetes mellitus, chronic kidney disease, alcohol use disorder, cirrhosis, and any prior GIB.

Phecode and CCS Based Mappings

One version of feature engineering used features from medical history that were mapped using the Clinical Classification System (CCS) from the Agency of Healthcare Research and Quality (AHRQ) and the Phecode mapping system.[37, 38] In order to reduce the number of features used by the final model, several dimensionality reduction techniques were used including principal component analysis (PCA), canonical correlation analysis (CCA), a variational autoencoder, and an embedding from a supervised feedforward neural network. For PCA and CCA, the dimension of the embeddings were chosen to represent at least 90% of the total variance. Details about the embedding schemes applied to CCS and Phecode mapped features can be found in Table A2.1 and Table A2.2.

Embedding Methods

The four embedding methods were applied to each of the sets of embedding input variables resulting in a total of 16 different embedding schemes.

Table A2.1:

Complete list embedding input variables. Previously described mappings include Phecodes (PHE) and the Clinical Classification System (CCS).

Embedding Input Variables	Number of Classes
PHE Codes	1,867
PHE Codes + Medication classes (separate embeddings)	2,283

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Embedding Input Variables	Number of Classes
CCS Codes	530
CCS Codes + Medication classes (separate embeddings)	946

Table A2.2:

Complete list of embedding Methods

Embedding Methods	Description
Principal Component Analysis (PCA)	Statistical technique used for dimensionality reduction that performs a linear transformation of variables and results in a new coordinate system ordered by the greatest amount of explained variation.
Canonical Correlation Analysis (CCA)	Statistical technique used for dimensionality reduction that identifies linear combinations of variables with maximum correlation.
Variational Autoencoder	Neural network architecture that uses two separate neural networks, an encoder and decoder, trained together to map variables into a non-linear latent space.
Supervised Neural Network Embedding	Neural network architecture that uses a feedforward neural network to map variables to the desired outcome, with the final layer used as the non-linear latent space.

Custom Selection of ICD-10 Codes + Medications based on LASSO

An alternate method for selecting the ICD-10 codes and medication classes to include in the model incorporated the correlation between different codes and medications and the outcome of interest. Specifically, L1 penalized regressions from the glmnet package in R were run with a variety of hyperparameters regressing the outcomes on all ICD-10 codes and medications. The use of the L1 penalty induced sparsity in the coefficients of the resulting regressions, and the non-zero coefficients of the top 20 regressions by cross validation error were used to create 20 custom selected subsets of the ICD-10 codes and medications in order to train the models on.

A3. Hyperparameters for Machine Learning and Deep Learning Estimators

Hyperparameters for the gradient boosted tree model were tuned using a randomized grid-search approach, where the grid contains the number of rounds (25, 50, 100), maximum depth (1 to 12), learning rate eta (0.001, 0.01, 0.05, 0.1, 0.2, 0.3), minimum loss reduction required to make a further partition on a leaf node (0, 1, 2, 3), fraction of features that are randomly sampled for tree splitting (0.4, 0.7, 1.0), minimum sum of instance weight needed in a child node (0.5, 1, 1.5), and fraction of samples randomly sampled to train each tree (0.5, 0.6, 0.7, 1).

Hyperparameters for the random forest model were also tuned using a randomized grid-search approach, where the grid contains the number of variables randomly selected at each split point (uniformly sampled between 0.2 to 0.6 times the number of variables), the minimum number of observations to follow strictly in the leaf nodes (1 to 15), and the proportion of training data used as the splitting dataset (0.51 to 0.75).

For the deep learning neural network model, we trained two feedforward neural networks: one with 2 layers and another with 5 layers with ReLU activation functions attached

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to a binary classifier to predict outcome. The weights were initialized using the Glorot initialization and trained using Adam optimizer to minimize the RMSE with dropout regularization on all layers to prevent overfitting. We used an adaptive cross validation with random grid search approach to tune the hyperparameters. The grid contains the number of units (8, 16, 32, 64, 128) and dropouts' rate (0.1, 0.2, 0.3, 0.4) for each layer, and the number of epochs (20, 30).

Parameters selected for the final XGBoost model: Learning rate (eta) = .05, Minimum split loss (gamma) = 0.0, Maximum tree depth (max_depth) = 3, Minimum sum of instance weight in a child (min_child_weight) = .5, Subsample ratio of training instances (subsample) = 0.7, Subsample ratio of columns when constructing each tree (colsample_bytree) = 1.

Table A3:

Performance of the top performing models (based on Test AUC) and Clinical Risk Scores

Data	Estimator	Embedding	Internal Validation		External Validation	
			AUC	Specificity at 99% sensitivity	AUC	Specificity at 99% sensitivity
PHE/meds	XGBoost	LASSO	0.909	0.424	0.923	0.379
ICD/meds	XGBoost	LASSO	0.909	0.437	0.923	0.323
CCS	XGBoost	PCA	0.909	0.408	0.923	0.341
CCS	XGBoost	LASSO	0.909	0.468	0.923	0.350
CCS/meds	XGBoost	LASSO	0.909	0.464	0.923	0.330
ICD	XGBoost	LASSO	0.909	0.454	0.922	0.294
PHE	XGBoost	LASSO	0.908	0.380	0.921	0.310
CCS/meds	XGBoost	PCA	0.911	0.424	0.921	0.359
CCS/meds	XGBoost	PCA	0.901	0.361	0.921	0.352
GBS Score		N/A			0.890	0.185
Oakland Score		N/A			0.891	0.117

A4 Calibration Details

While we train our model with accuracy as the loss metric, we emphasize that the raw risk score predictions of the model are not calibrated to be interpreted as conditional probabilities. In order to be useful for clinical problems, one must calibrate the outputted risk scores of the model against a threshold and compare to this threshold when determining whether or not a specific patient is high risk or not. Based on our validation data, we found the 99% sensitivity threshold for our model to be .105, meaning that any patients with predicted risk scores greater than .105 are at high risk of needing a hospital-based intervention, and patients with predicted risk scores lower than .105 are at low risk of needing a hospital-based intervention. We can also run an auxiliary regression to calibrate our risk scores to be conditional probabilities of needing a hospital-based intervention.

It is also of interest to compare the calibration of our ML model scores with that of traditional risk scores, such as the Glasgow-Blatchford Score. Specifically, we can test the

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Brier Scores of our model's calibrated conditional probabilities with those of the Glasgow-Blatchford Score. Our selected model, XGboost, ended up with a Brier score of 0.1056, compared to a Brier score of 0.1302 for the Glasgow-Blatchford Score and a score of 0.1256 for the Oakland score.

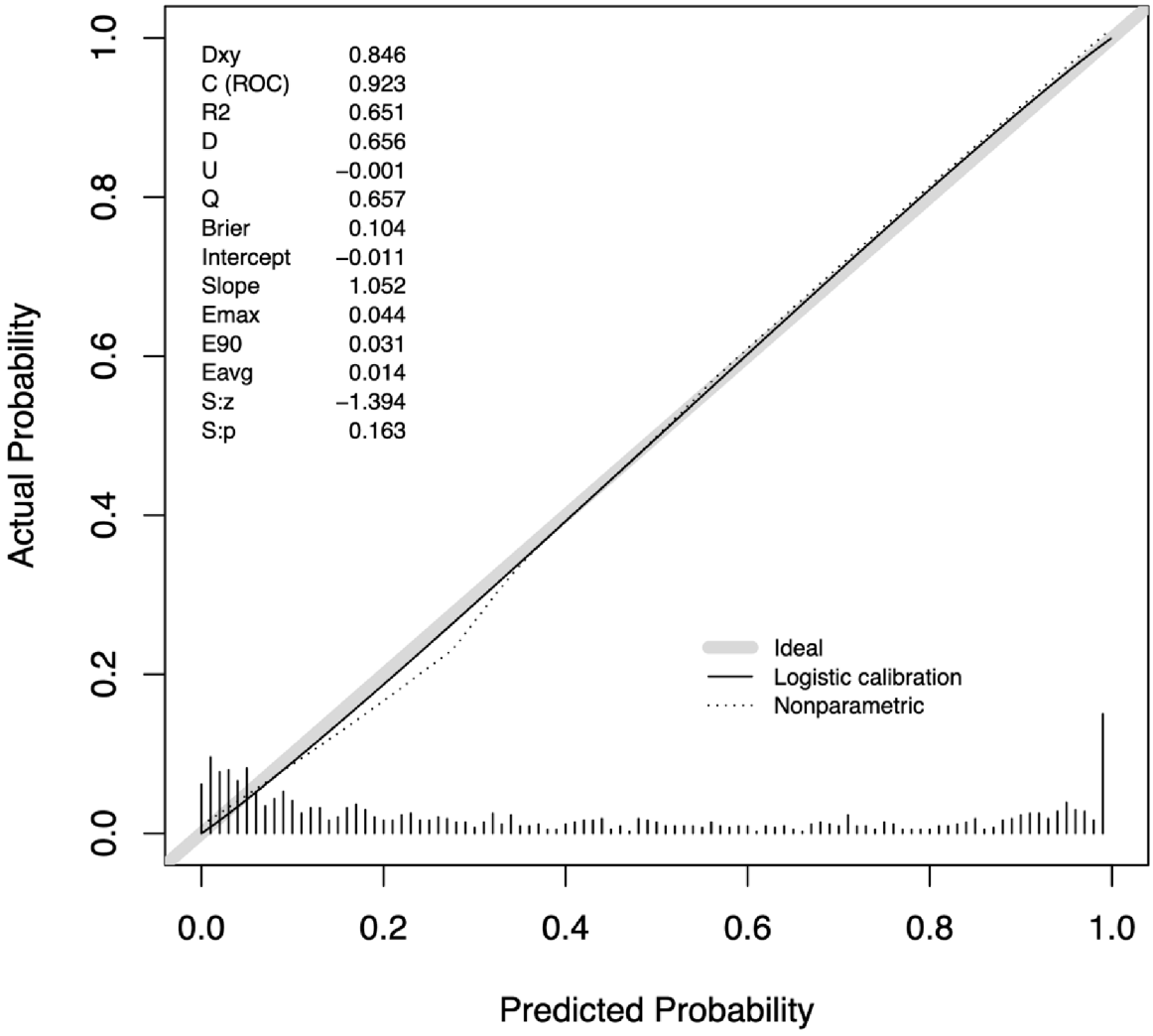


Figure A4.1:
Calibration of conditional probabilities predicted by the machine learning model.

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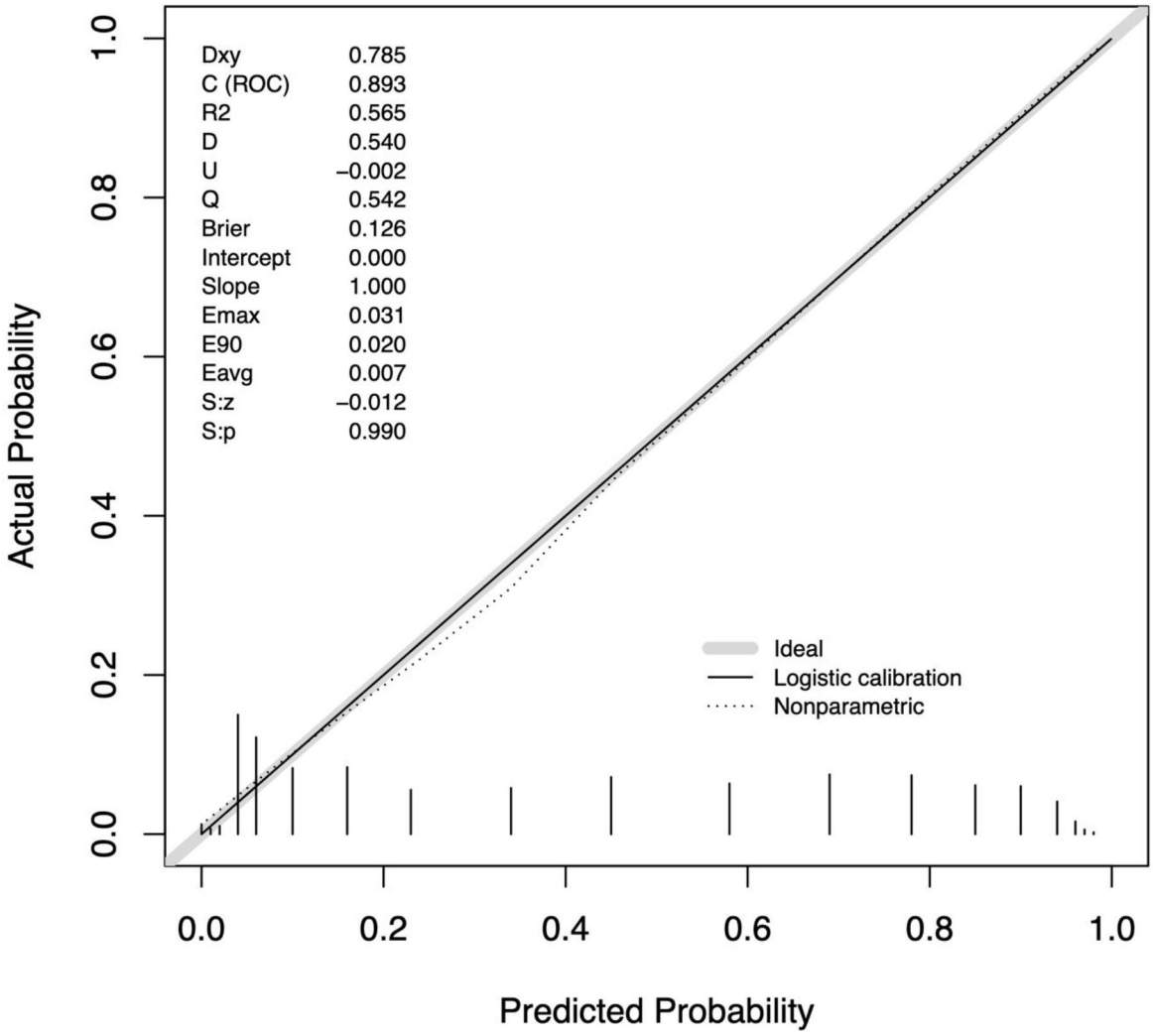


Figure A4.2:
Calibration of conditional probabilities predicted by the Glasgow-Blatchford Score.

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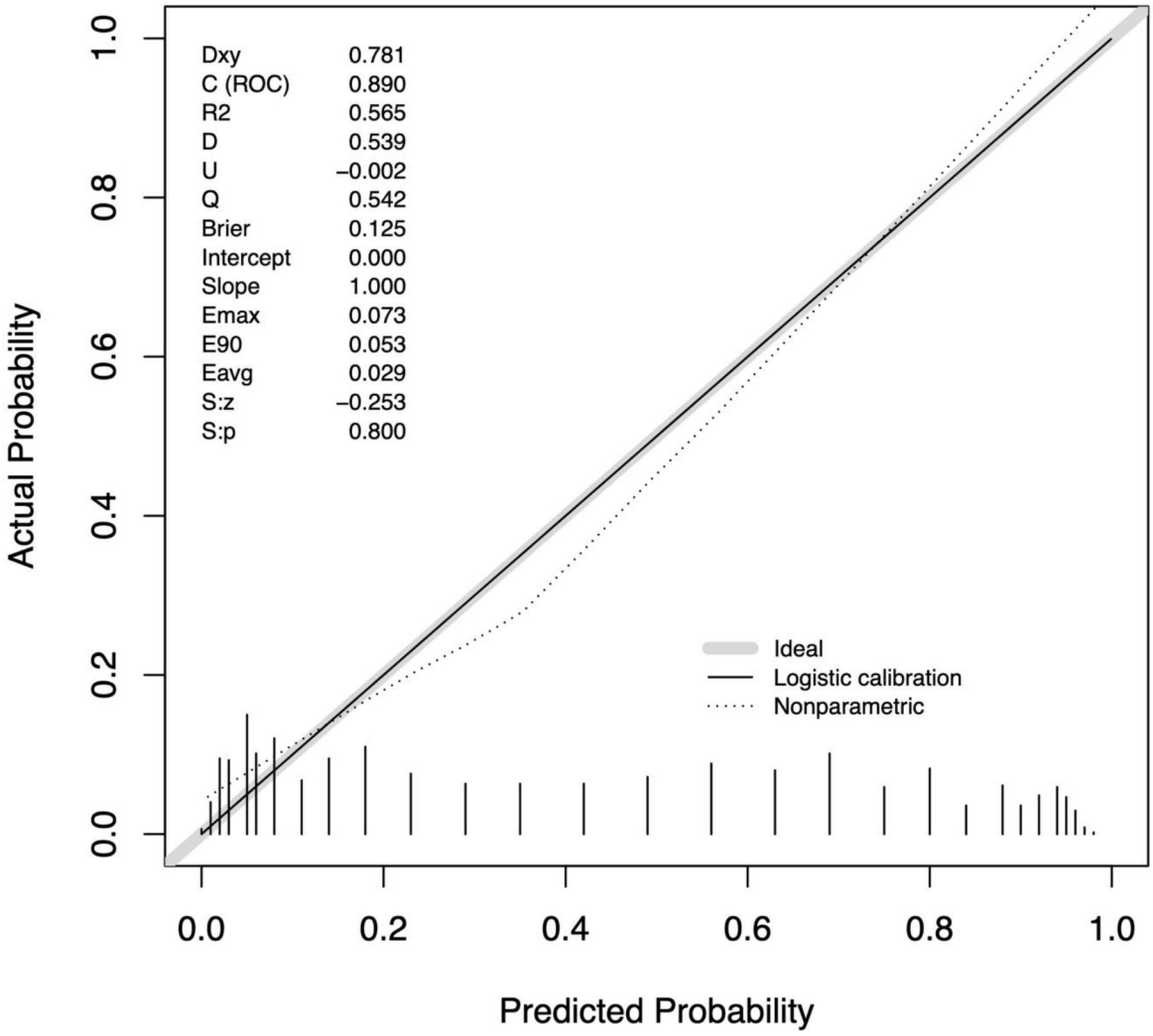


Figure A4.3:
Calibration of conditional probabilities predicted by the Oakland score.

A5 Components for Glasgow-Blatchford Score and Oakland Score

The Glasgow-Blatchford Score ranges from 0 to 23 and is comprised of blood urea nitrogen (mmol/L), hemoglobin (g/dL), systolic blood pressure, presence of tachycardia, melena, syncope, hepatic disease, and cardiac failure.[28] The points assigned to cutoffs for each of the variables is as follows. For blood urea nitrogen, ≥70 mg/dL contributes 6 points, ≥28 mg/dL and <70 mg/dL contributes 4 points, ≥22.4 mg/dL and <28 mg/dL contributes 3 points, ≥18.2 mg/dL and <22.3 mg/dL contributes 2 points, and <18.2 mg/dL 0 points. Hemoglobin contributes points according to biological sex: for men, hemoglobin ≥13 g/dL contributes 0 points, ≥12 g/dL and <13 g/dL contributes 1 point, ≥10 g/dL and <12 g/dL contributes 3 points, and <10 g/dL contributes 6 points; for women, hemoglobin ≥12 g/dL contributes 0 points, ≥10 g/dL and <12 g/dL contributes 1 point and <10 g/dL contributes 6 points. For systolic blood pressure, >109 mmHg contributes 0 points, 100–109 mmHg contributes 1 point, 90–99 mmHg contributes 2 points, and <90 mmHg contributes 3 points.

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If the pulse is ≥100 beats per minutes, it contributes 1 point, otherwise 0 points. Presence of melena contributes 1 point, presentation with syncopal symptoms contributes 2 points, hepatic disease contributes 2 points, and cardiac failure contributes 2 points.

The modified Oakland Score ranges from 0 to 34 and is comprised of age, biological sex, previous hospital admission with lower gastrointestinal bleeding, heart rate, systolic blood pressure, and hemoglobin.[18] For age, ≤39 years old contributes 0 points, 40–69 years old contributes 1 point, and ≥70 years old contributes 2 points. If the patient is male, the biological sex contributes 1 point. Previous hospital admission with LGIB contributes 1 point. For heart rate, ≤69 beats per minute contributes 0 points, 70–89 beats per minute contributes 1 point, 90–109 beats per minute contributes 2 points, and ≥110 beats per minute contributes 3 points. For systolic blood pressure, 50–89 mmHg contributes 5 points, 90–119 mmHg contributes 4 points, 120–129 mmHg contributes 3 points, 130–159 mmHg contributes 2 points, and ≥160 mmHg contributes 0 points. For hemoglobin levels, 3.6–6.9 g/dL contributes 22 points, 7.0–8.9 g/dL contributes 17 points, 9.0–10.9 g/dL contributes 13 points, 11.0–12.9 g/dL contributes 8 points, 13.0–15.9 g/dL contributes 4 points, and ≥16.0 g/dL contributes 0 points.

A6 Role and Definitions of Interpretable Methods

Clinicians and researchers ideally want to understand which factors play a significant role and how they interact with each other to lead to the predicted outcome. In order to help engender trust in the accuracy and stability of our risk model, as well as allow practitioners to examine the internal relationships learned by the model, we make use of a number of interpretability methods to explain the predictions of the model.

Partial Dependence Plots (PDPs) offer a global view on how a single predictor affects the outcome while averaging out the effects of all other predictors. In simpler terms, if we're considering a factor like age in the risk prediction for gastrointestinal bleeding, a PDP would show us the average effect of age on the risk, while holding all other factors constant. This visualization gives a broad overview and can help in understanding general trends or patterns. However, PDPs can sometimes be misleading if there are interactions between variables that are not considered.

Accumulated Local Expectation Plots (ALEs), while conceptually similar to PDPs, take a slightly different approach. Instead of looking at the average effect of a predictor, they measure how changes in a predictor's value leads to changes in the model's prediction by focusing on the difference from an average prediction. ALEs are particularly valuable because they automatically account for interactions between the predictor in question and other predictors in the model, offering a more granular insight compared to PDPs.

While both PDPs and ALEs provide a global perspective on predictor influence, Individual Conditional Expectation Plots (ICE) plots dive deep into individual predictions. Each line in an ICE plot represents how a single instance (or patient, in our case) will react to changes in a particular predictor, holding all others constant. For the gastrointestinal bleeding risk model, ICEs could show how the risk prediction for a specific patient changes as their age

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or any other factor varies. This individual-level understanding is crucial for personalized medicine and tailoring interventions to specific patients based on their unique profiles.

When clinicians aim to understand the inner logic of predictive risk models for gastrointestinal bleeding, understanding the role and influence of individual predictors is crucial. PDPs, ALEs, and ICEs provide different lenses through which we can interpret these roles, each with its own set of advantages and considerations. By employing these tools, clinicians and researchers can achieve a deeper understanding of their models and make more informed decisions for patient care.

A7 Interpretability Dashboard

While the use of black box machine learning models for risk assessment offers many upsides for practitioners, including increased predictive performance (as compared to the Glasgow-Blatchford Score), the ability to transfer models across various hospital systems, and the ability to incorporate arbitrary structured and unstructured predictive features into a risk model, they in turn have several downsides, chiefly, the ease of interpretation of the model logic. Decision tree algorithms often contain hundreds of partitions of the input space along alternating features leading to many decision paths contributing to a specific prediction. When they are ensembled into a random forest or gradient boosting ensemble, the complexity of interpreting the model only increases. Interpreting machine learning models is important beyond just understanding the prediction logic. Interpretability metrics offer practitioners the ability to audit the internals of the model, learn from feature relationships selected by the model, learn from these relationships in creating future models, and show the stability of the output of the model to gain trust from stakeholders.

A8 Disparity Analysis

As a post-hoc analysis, we analyze the performance of our chosen model by patient characteristics, namely, sex, ethnicity, and race. We find no difference in performance, as measured by AUC on the test set, across the different patient characteristics.

Table A8.1:

Performance of machine learning model across subgroups defined by race, ethnicity and biological sex.

Category		N	AUC	Percent of patients who require hospital-based intervention	Number of patients below 99% sensitivity threshold	True Negatives	Patients correctly identified as very-low-risk (% of total very-low-risk patients)
Sex							
	Female	409	0.934	38.6	82	81	98.8
	Male	517	0.917	40.8	132	130	98.5
Race							
	Black/African-American	180	0.918	30.6	60	60	100

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Category		N	AUC	Percent of patients who require hospital-based intervention	Number of patients below 99% sensitivity threshold	True Negatives	Patients correctly identified as very-low-risk (% of total very-low-risk patients)
	White	555	0.914	42.5	95	93	97.9
	Other/ unknown	191	0.951	40.8	59	58	98.3
Ethnicity							
	Hispanic	215	0.960	36.3	68	68	100
	Not Hispanic	708	0.910	40.8	146	143	97.9

Table A8.2:

Performance of machine learning model across subgroups defined by signs or symptoms of acute overt gastrointestinal bleeding: hematemesis, hematochezia, and melena.

Category		N	AUC	Percent of patients who require hospital-based intervention	Number of patients below 99% sensitivity threshold	True Negatives	Patients correctly identified as very-low-risk (% of total very-low-risk patients)
Hematemesis							
	Hematemesis with or without hematochezia or melena	259	0.915	38.7	44	44	100
	Hematochezia and/or melena only	667	0.926	42.9	170	167	98.2
Hematochezia							
	Hematochezia with or without hematemesis or melena	517	0.934	38.1	132	130	98.5
	Hematemesis and/or melena only	409	0.909	42.1	82	81	98.8
Melena							
	Melena with or without hematemesis or hematochezia	657	0.908	40.9	59	58	98.3
	Hematemesis and/or hematochezia only	269	0.931	39.4	155	153	98.7

A9 Error Analysis

Using the 99% sensitivity (very-low-risk) threshold, we identified the false positives (cases where patients were identified by the machine learning model as not very low risk who did not in fact need hospital-based intervention) and compared their characteristics to the true negatives. We display results of the base set variables corresponding to a p-value < 0.05 (Table 9.1). We also plotted the distributions of the significant base set variables of the correctly predicted values vs. false positive ones (Figure 9.1). Among these, the most significant predictors ($p < 0.001$) were systolic blood pressure, hemoglobin, and hematocrit.

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The false positive values of the two variables typically fell within their normal ranges, which is an expected finding since these instances were false positives identified at a very low threshold. The bottom right histogram of Figure 9.1 reveals that most of the false positive predictions were marginally above our very-low-risk threshold. This suggests that these cases might have been correctly classified given a slightly higher threshold for determining very low risk.

Table 9.1:

Comparison of base variable characteristics of patients misclassified as not-very-low risk who did not require hospital intervention (false positives) versus true negatives.

Variable	Very Low Risk Patients with Correct Prediction N=577	Very Low Risk Patients with Incorrect Prediction N=346	Difference	P-value
Age at admission	67.6	64.1	3.4	< 0.01
Systolic BP	139	131	8	< 0.001
Diastolic BP	75	73	2	< 0.05
Bicarbonate (CO2)	25	24	−1	< 0.05
Blood Urea Nitrogen (BUN)	24.5	28.2	−3.7	< 0.01
Hemoglobin	11.6	10.5	1.1	< 0.001
Hematocrit	35.4	32.0	3.4	< 0.001
Mean Corpuscular Hemoglobin Concentration (MCHC)	32.8	32.5	0.3	< 0.05
Alkaline Phosphatase (ALP)	110.4	97.7	12.7	< 0.05
Albumin	3.8	3.6	0.15	< 0.01
Shock Index	0.66	0.70	−0.04	< 0.01

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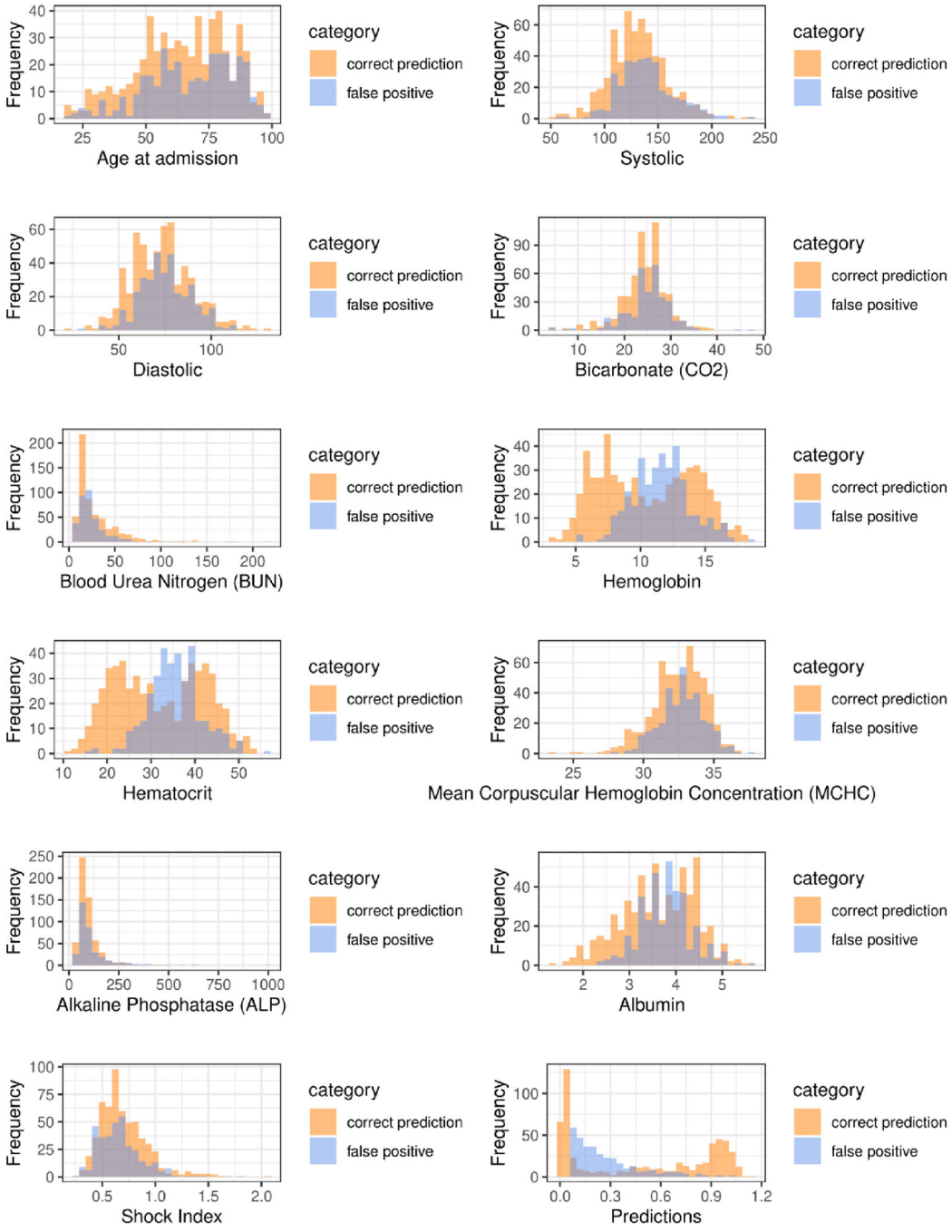


Figure 9.1:

Superimposed histograms displaying data distributions of 9 variables with differences that were significant by P<0.05 between the false positives and true negatives across the base variable characteristics reported above.

Abbreviations:

AI-CDS	Sartificial intelligence clinical decision support system

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AUC	area under receiver operating characteristic curve
EHR	electronic health record
GBS	Glasgow-Blatchford Score
GIB	gastrointestinal bleeding

Literature Cited

1. Peery AF, Crockett SD, Murphy CC, et al. Burden and Cost of Gastrointestinal, Liver, and Pancreatic Diseases in the United States: Update 2021. Gastroenterology 2022;162:621–644. [PubMed: 34678215]
2. Laine L Upper Gastrointestinal Bleeding Due to a Peptic Ulcer. N Engl J Med 2016;375:1198.
3. Gralnek IM, Neeman Z, Strate LL. Acute Lower Gastrointestinal Bleeding. New England Journal of Medicine 2017;376:1054–1063. [PubMed: 28296600]
4. Laine L, Barkun AN, Saltzman JR, et al. ACG Clinical Guideline: Upper Gastrointestinal and Ulcer Bleeding. Am J Gastroenterol 2021;116:899–917. [PubMed: 33929377]
5. Gralnek IM, Stanley AJ, Morris AJ, et al. Endoscopic diagnosis and management of nonvariceal upper gastrointestinal hemorrhage (NVUGIH): European Society of Gastrointestinal Endoscopy (ESGE) Guideline - Update 2021. Endoscopy 2021;53:300–332. [PubMed: 33567467]
6. Oakland K, Chadwick G, East JE, et al. Diagnosis and management of acute lower gastrointestinal bleeding: guidelines from the British Society of Gastroenterology. Gut 2019;68:776–789. [PubMed: 30792244]
7. Stanley AJ, Ashley D, Dalton HR, et al. Outpatient management of patients with low-risk upper-gastrointestinal haemorrhage: multicentre validation and prospective evaluation. Lancet 2008;373:42–7. [PubMed: 19091393]
8. Shung DL, Lin JK, Laine L. Achieving Value by Risk Stratification with Machine Learning Model or Clinical Risk Score in Acute Upper Gastrointestinal Bleeding: A Cost Minimization Analysis. Am J Gastroenterol 2023.
9. Liang PS, Saltzman JR. A National Survey on the Initial Management of Upper Gastrointestinal Bleeding. Journal of Clinical Gastroenterology 2014;48.
10. Hayes SM, Murray S, Dupuis M, et al. Barriers to the Implementation of Practice Guidelines in Managing Patients with Nonvariceal Upper Gastrointestinal Bleeding: A Qualitative Approach. Canadian Journal of Gastroenterology 2010;24:878135.
11. Laine L. Risk Assessment Tools for Gastrointestinal Bleeding. Clinical Gastroenterology and Hepatology 2016;14:1571–1573. [PubMed: 27521511]
12. Stanley AJ, Laine L, Dalton HR, et al. Comparison of risk scoring systems for patients presenting with upper gastrointestinal bleeding: international multicentre prospective study. Bmj 2017;356:i6432. [PubMed: 28053181]
13. Laursen SB, Hansen JM, Schaffalitzky de Muckadell OB. The Glasgow Blatchford score is the most accurate assessment of patients with upper gastrointestinal hemorrhage. Clinical Gastroenterology and Hepatology 2012;10:1130–1135.e1. [PubMed: 22801061]
14. Ramaekers R, Mukarram M, Smith CA, et al. The Predictive Value of Preendoscopic Risk Scores to Predict Adverse Outcomes in Emergency Department Patients With Upper Gastrointestinal Bleeding: A Systematic Review. Acad Emerg Med 2016;23:1218–1227. [PubMed: 27640399]
15. Laursen SB, Dalton HR, Murray IA, et al. Performance of new thresholds of the Glasgow Blatchford score in managing patients with upper gastrointestinal bleeding. Clin Gastroenterol Hepatol 2015;13:115–21 e2. [PubMed: 25058843]
16. Oakland K, Jairath V, Uberoi R, et al. Derivation and validation of a novel risk score for safe discharge after acute lower gastrointestinal bleeding: a modelling study. Lancet Gastroenterol Hepatol 2017;2:635–643. [PubMed: 28651935]

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17. Almaghrabi M, Gandhi M, Guizzetti L, et al. Comparison of Risk Scores for Lower Gastrointestinal Bleeding: A Systematic Review and Meta-analysis. JAMA Netw Open 2022;5:e2214253. [PubMed: 35622365]
18. Oakland K, Kothiwale S, Forehand T, et al. External Validation of the Oakland Score to Assess Safe Hospital Discharge Among Adult Patients With Acute Lower Gastrointestinal Bleeding in the US. JAMA Network Open 2020;3:e209630–e209630. [PubMed: 32633766]
19. Shung DL, Au B, Taylor RA, et al. Validation of a Machine Learning Model That Outperforms Clinical Risk Scoring Systems for Upper Gastrointestinal Bleeding. Gastroenterology 2020;158:160–167. [PubMed: 31562847]
20. Shung D, Simonov M, Gentry M, et al. Machine Learning to Predict Outcomes in Patients with Acute Gastrointestinal Bleeding: A Systematic Review. Dig Dis Sci 2019.
21. Goldstein BA, Navar AM, Pencina MJ, et al. Opportunities and challenges in developing risk prediction models with electronic health records data: a systematic review. Journal of the American Medical Informatics Association : JAMIA 2017;24:198–208. [PubMed: 27189013]
22. Yang C, Kors JA, Ioannou S, et al. Trends in the conduct and reporting of clinical prediction model development and validation: a systematic review. J Am Med Inform Assoc 2022;29:983–989. [PubMed: 35045179]
23. Rodriguez NJ, Zheng N, Mezzacappa C, et al. DISPARITIES IN ACCESS TO ENDOSCOPY FOR PATIENTS WITH UPPER GASTROINTESTINAL BLEEDING PRESENTING TO EMERGENCY DEPARTMENTS. Gastroenterology 2022.
24. Liao KP, Cai T, Savova GK, et al. Development of phenotype algorithms using electronic medical records and incorporating natural language processing. BMJ (Clinical research ed.) 2015;350:h1885–h1885.
25. Shung D, Tsay C, Laine L, et al. Early identification of patients with acute gastrointestinal bleeding using natural language processing and decision rules. Journal of Gastroenterology and Hepatology 2020;n/a.
26. Shung D, Huang J, Castro E, et al. Neural network predicts need for red blood cell transfusion for patients with acute gastrointestinal bleeding admitted to the intensive care unit. Sci Rep 2021;11:8827. [PubMed: 33893364]
27. Deshmukh F, Merchant SS. Explainable Machine Learning Model for Predicting GI Bleed Mortality in the Intensive Care Unit. Am J Gastroenterol 2020;115:1657–1668. [PubMed: 32341266]
28. Laursen SB, Oakland K, Laine L, et al. ABC score: a new risk score that accurately predicts mortality in acute upper and lower gastrointestinal bleeding: an international multicentre study. Gut 2020:gutjnl-2019–320002.
29. Marmo R, Soncini M, Bucci C, et al. Derivation and validation of Re.Co.De death score risk in patients with acute nonvariceal upper GI bleeding. Gastrointest Endosc 2022;96:36–43.e8. [PubMed: 35150665]
30. Levi R, Carli F, Arévalo AR, et al. Artificial intelligence-based prediction of transfusion in the intensive care unit in patients with gastrointestinal bleeding. BMJ health & care informatics 2021;28:e100245.
31. Friedman J, Hastie T, Tibshirani R. Regularization Paths for Generalized Linear Models via Coordinate Descent. J Stat Softw 2010;33:1–22. [PubMed: 20808728]
32. Chen T, Guestrin C. XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. San Francisco, California, USA: Association for Computing Machinery, 2016:785–794.
33. Künzel SR, Saarinen TF, Liu EW, et al. Linear Aggregation in Tree-Based Estimators. Journal of Computational and Graphical Statistics 2022;31:917–934.
34. Kuhn M. Building Predictive Models in R Using the caret Package. Journal of Statistical Software 2008;28:1 – 26. [PubMed: 27774042]
35. Blatchford O, Murray WR, Blatchford M. A risk score to predict need for treatment for upper-gastrointestinal haemorrhage. Lancet 2000;356:1318–21. [PubMed: 11073021]
36. Steyerberg EW. Validation in prediction research: the waste by data splitting. J Clin Epidemiol 2018;103:131–133. [PubMed: 30063954]

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37. Sutton RT, Pincock D, Baumgart DC, et al. An overview of clinical decision support systems: benefits, risks, and strategies for success. NPJ Digit Med 2020;3:17. [PubMed: 32047862]

38. Henry KE, Kornfield R, Sridharan A, et al. Human-machine teaming is key to AI adoption: clinicians' experiences with a deployed machine learning system. NPJ Digit Med 2022;5:97. [PubMed: 35864312]

39. Tonekaboni S, Joshi S, McCradden MD, et al. What clinicians want: contextualizing explainable machine learning for clinical end use, In Machine learning for healthcare conference, PMLR, 2019.

40. Greenwell BM, Boehmke BC, McCarthy AJ. A simple and effective model-based variable importance measure. arXiv preprint arXiv:1805.04755 2018.

41. Goldstein A, Kapelner A, Bleich J, et al. Peeking inside the black box: Visualizing statistical learning with plots of individual conditional expectation. journal of Computational and Graphical Statistics 2015;24:44–65.

42. Chishtie J, Sapiro N, Wiebe N, et al. Use of Epic Electronic Health Record System for Health Care Research: Scoping Review. J Med Internet Res 2023;25:e51003. [PubMed: 38100185]

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What You Need to Know

BACKGROUND AND CONTEXT

An electronic health record (EHR)-based machine learning model may outperform existing guideline-recommended risk stratification scores in identifying very-low-risk patients with acute gastrointestinal bleeding.

NEW FINDINGS

We developed an EHR-based machine learning model that outperforms the Glasgow-Blatchford Score and Oakland Score in identifying patients presenting with acute gastrointestinal bleeding who met a composite endpoint of hospital-based intervention or death. At a threshold of 99% sensitivity for the composite endpoint, the machine learning model identified substantially more very-low-risk patients than the current gold standard Glasgow-Blatchford and Oakland Scores.

LIMITATIONS

This model is limited to complete cases, requires further validation in other populations, and requires real-world implementation studies.

CLINICAL RESEARCH RELEVANCE

Our model could increase identification of very-low-risk patients who can be safely discharged from the emergency department for outpatient management. This research demonstrates the utility of using electronic health record data for clinical risk predictions.

BASIC RESEARCH RELEVANCE

Our machine learning model trained on electronic health record data outperforms existing clinical risk scores.

LAY SUMMARY

Electronic health record-based machine learning models can provide automated, interpretable risk stratification for acute gastrointestinal bleeding to identify very-low-risk patients who could benefit from outpatient management.

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Figure 1:

Artificial Intelligence-Clinical Decision Support System – an Interactive Interpretable Machine Learning Dashboard. The inputs on the left show demographics, vital signs, and other input categories such as laboratory values, medications, and past medical history. The Top pane displays a patient that is not very low risk (probability is above the pre-set threshold for the high sensitivity 99%) and Bottom pane a very-low-risk patient. This is defined by the model probability output less than 6.3%, which is the output that corresponds to the 99% sensitivity threshold for very low risk patients. The top three variables that

affect risk are displayed in graphical form, with how the risk changes across the different thresholds of the variable (e.g. as hemoglobin increases, risk of hospital-based intervention decreases)

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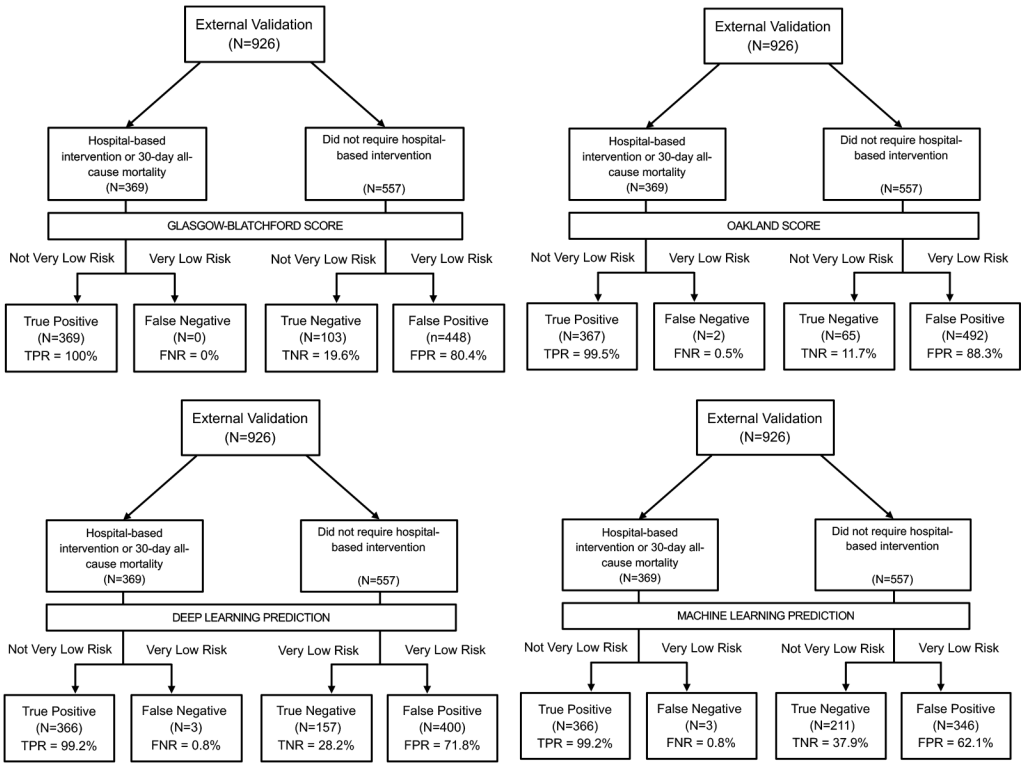


Figure 2:

Flow diagram for the selected machine learning model (top left), deep learning model (top right), GBS (bottom left), and Oakland score (bottom right) with the respective predicted classifications, false positive, false negative, true positive, and true negative rates.

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Table 1:

Baseline demographic, clinical outcomes (red blood cell transfusion, hemostatic intervention, all-cause 30-day mortality), initial measured laboratory values, medications at presentation and co-morbidities defined by ICD-10 codes for training and external validation cohorts. Values are reported as proportions for count variables and mean with 95% standard deviation for continuous variables.

	Training Set (N=2546)	Internal Validation (N=850)	External Validation (N=926)	Difference (Training and External Validation)
Demographics				
Age at Admission	64.8 (18.4)	65.6 (17.8)	65.5 (17.8)	0.6 (−0.7 to 2.0)
Sex				
Female	1179 (46.3%)	404 (47.5%)	409 (44.2%)	2.1 (−1.6 to 5.9)
Male	1367 (53.7%)	446 (52.5%)	517 (55.8%)	
Outcomes				
30-day Mortality	141 (5.5%)	38 (4.5%)	56 (6.0%)	0.5 (−1.3 to 2.3)
Red Blood Cell Transfusion	857 (33.7%)	287 (33.8%)	319 (34.4%)	0.8 (−2.8 to 4.4)
Endoscopic Hemostatic Intervention	196 (7.7%)	68 (8.0%)	93 (10.0%)	2.3 (0.1 to 4.5)
Surgical Intervention	120 (4.7%)	38 (4.5%)	45 (4.9%)	0.1 (−1.5 to 1.8)
Interventional Radiologic Intervention	32 (1.3%)	12 (1.4%)	1 (0.1%)	−1.1 (−1.6 to −0.7)
Any Hemostatic Intervention	272 (10.7%)	91 (10.7%)	111 (12.0%)	1.3 (−1.1 to 3.7)
Composite Outcome	972 (38.2%)	326 (38.4%)	369 (39.8%)	1.7 (−2.0 to 5.4)
Initial Lab Measurement	**Mean (SD)**		**Mean (SD)**	
Sodium (mmol/L)	139.1 (4.3)	139.3 (4.2)	139.2 (4.4)	0.2 (−0.2 to 0.5)
Potassium (mEq/L)	4.2 (0.7)	4.3 (0.7)	4.3 (0.6)	0.1 (0.0 to 0.1)
Chloride (mEq/L)	100.5 (5.7)	100.9 (5.7)	102.7 (5.4)	2.2 (1.8 to 2.7)
Bicarbonate (mEq/L)	22.7 (4.0)	22.5 (4.1)	24.5 (4.7)	1.9 (1.5 to 2.2)
Blood Urea Nitrogen (mg/dL)	28.4 (22.8)	28.8 (22.7)	26.8 (20.9)	−1.6 (−3.2 to −0.1)
Creatinine (mg/dL)	1.4 (1.6)	1.5 (1.6)	1.3 (1.5)	−0.1 (−0.2 to 0.0)
BUN/Creatinine Ratio	23.1 (13.3)	23.0 (13.0)	23.8 (13.7)	0.7 (−0.3 to 1.7)
International Normalized Ratio (INR)	1.4 (1.0)	1.4 (1.2)	1.4 (1.3)	0.1 (0.0 to 0.2)
White Blood Cell count (WBCs per mL)	10.0 (6.0)	9.5 (5.8)	9.6 (4.8)	−0.5 (−0.8 to −0.1)
Hemoglobin (g/dL)	11.1 (3.0)	11.0 (3.0)	10.9 (3.2)	−0.1 (−0.4 to 0.1)
Platelet count (Platelets per mcL)	244.9 (123.0)	232.8 (105.8)	234.3 (105.9)	−10.6 (−18.9 to −2.2)
Hematocrit (HCT)	34.4 (8.9)	34.3 (8.8)	33.3 (8.9)	−1.1 (−1.8 to −0.5)
Mean Corpuscular Volume (MCV, fl)	89.9 (8.6)	90.5 (8.0)	90.2 (7.9)	0.3 (−0.3 to 0.9)
Mean Corpuscular Hemoglobin (MCH, pg per cell)	29.5 (3.4)	29.6 (3.2)	29.4 (3.2)	0.0 (−0.3 to 0.2)
Mean Corpuscular Hemoglobin Concentration (MCHC, g/dL)	32.7 (1.7)	32.7 (1.6)	32.6 (1.8)	−0.1 (−0.3 to 0.0)
Alanine Aminotransferase (ALT, U/L)	28.4 (46.7)	28.6 (40.0)	43.9 (88.6)	15.5 (9.5 to 21.5)

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	Training Set (N=2546)	Internal Validation (N=850)	External Validation (N=926)	Difference (Training and External Validation)
Aspartate Aminotransferase (AST, U/L)	42.6 (116.2)	42.6 (77.6)	51.5 (143.2)	8.8 (1.5 to 19.1)
Alkaline Phosphatase (U/L)	105.3 (106.1)	99.1 (71.4)	102.4 (73.6)	−2.9 (−9.2 to 3.4)
Total Bilirubin (mg/dL)	1.0 (2.9)	0.9 (2.3)	1.0 (1.9)	0.0 (−0.2 to 0.1)
Albumin (g/dL)	3.7 (0.6)	3.6 (0.7)	3.7 (0.7)	0.0 (0.0 to 0.1)
Initial Vitals Measurement	**Mean (SD)**		**Mean (SD)**	
Heart Rate (beats per minute)	89.0 (20.5)	88.1 (19.9)	88.2 (19.1)	−0.8 (−2.2 to 0.7)
Oxygen Saturation (%)	97.5 (2.8)	97.3 (2.9)	97.2 (4.9)	−0.3 (−0.6 to 0.1)
Respiratory Rate (breaths per minute)	18.2 (3.6)	18.3 (4.0)	18.8 (2.7)	0.7 (0.4 to 0.9)
Temperature (degrees Fahrenheit)	98.0 (1.0)	97.9 (2.0)	97.7 (1.1)	−0.3 (−0.4 to −0.3)
Systolic Blood Pressure (mmHg)	128.1 (25.5)	127.8 (25.6)	134.2 (26.3)	6.1 (4.1 to 8.1)
Diastolic Blood Pressure (mmHg)	72.8 (17.2)	72.7 (17.7)	73.9 (15.2)	1.0 (−0.1 to 2.2)
Mean Arterial Pressure (MAP, mmHg)	91.2 (18.1)	91.0 (18.4)	94.1 (17.4)	3.0 (1.6 to 4.3)
Nursing Assessment	**Proportion**		**Proportion**	
Glasgow Coma Score – Eye Component				
Unmeasured	1196 (47.0%)	408 (48.0%)	511 (55.2%)	8.2 (4.5 to 11.9)
Normal	1287 (50.5%)	418 (49.2%)	391 (42.2%)	−8.3 (−12.1 to −4.6)
Abnormal	63 (2.5%)	24 (2.8%)	24 (2.6%)	0.1 (−1.1 to 1.3)
Glasgow Coma Score – Verbal Component				
Unmeasured	1198 (47.1%)	409 (48.1%)	510 (55.1%)	8.0 (4.3 to 11.8)
Normal	1176 (46.2%)	376 (44.2%)	361 (39.0%)	−7.2 (−10.9 to −3.5)
Abnormal	172 (6.8%)	65 (7.6%)	55 (5.9%)	−0.8 (−2.6 to 0.1)
Glasgow Coma Score – Motor Component				
Unmeasured	1199 (47.1%)	409 (47.1%)	513 (55.4%)	8.3 (4.6 to 12.1)
Normal	1280 (50.3%)	417 (49.1%)	391 (42.2%)	−8.1 (−11.8 to −4.3)
Abnormal	67 (2.6%)	24 (2.8%)	22 (2.4%)	−0.3 (−1.4 to 0.9)
Shock Index	0.7 (0.2)	0.7 (0.2)	0.7 (0.2)	0.0 (−0.1 to 0.0)
Past Medical History				
Charlson Comorbidity Index				
0	1132 (44.5%)	363 (42.7%)	425 (45.9%)	1.4 (−2.3 to 5.2)
1	642 (25.2%)	211 (24.8%)	220 (23.8%)	−1.5 (−4.7 to 1.8)
2	359 (14.1%)	140 (16.5%)	140 (15.1%)	1.0 (−1.7 to 3.7)
3+	413 (16.2%)	136 (16.0%)	141 (15.2%)	−1.0 (−3.7 to 1.7)
Stroke or Transient Ischemic Attack	67 (2.6%)	29 (3.4%)	25 (2.7%)	0.1 (−1.1 to 1.3)
Deep Venous Thrombosis	138 (5.4%)	46 (5.4%)	47 (5.1%)	−0.3 (−2.0 to 1.3)
Pulmonary Embolism	98 (3.8%)	35 (4.1%)	34 (3.7%)	−0.2 (−1.6 to 1.1)
Atrial Fibrillation	476 (14.5%)	108 (12.7%)	141 (15.2%)	0.8 (−1.9 to 3.5)
Upper Gastrointestinal Bleeding	53 (2.1%)	11 (1.3%)	17 (1.8%)	−0.2 (−1.3 to 0.8)

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	Training Set (N=2546)	Internal Validation (N=850)	External Validation (N=926)	Difference (Training and External Validation)
Lower Gastrointestinal Bleeding	243 (9.5%)	82 (9.6%)	85 (9.2%)	−0.4 (−2.5 to 1.8)
Unspecified Gastrointestinal Bleeding	192 (7.5%)	66 (7.8%)	55 (5.9%)	−1.6 (−3.4 to 0.2)
Peptic Ulcer Disease	48 (1.9%)	20 (2.4%)	29 (3.1%)	1.2 (0.0 to 2.5)
History of Helicobacter Pylori Infection	4 (0.2%)	4 (0.1%)	9 (1.0%)	0.8 (0.2 to 1.5)
Coronary Artery Disease	459 (18.0%)	144 (16.9%)	161 (17.4%)	−0.6 (−3.5 to 2.2)
Heart Failure	286 (11.2%)	118 (13.9%)	122 (13.2%)	1.9 (−0.6 to 4.4)
Hypertension	1486 (58.4%)	525 (61.8%)	612 (66.1%)	7.7 (4.1 to 11.3)
Type 2 Diabetes Melllitus	579 (22.7%)	200 (23.5%)	246 (26.6%)	3.8 (0.5 to 7.1)
Chronic Kidney Disease	274 (10.8%)	108(12.7%)	69 (7.5%)	−3.3 (−5.4 to −1.2)
Alcohol Use Disorder	175 (6.9%)	54 (6.4%)	55 (5.9%)	−0.9 (−2.7 to 0.9)
Cirrhosis	219 (8.6%)	94 (11.1%)	64 (6.9%)	−1.7 (−3.7 to 0.3)
Medications prior to admission				
Aspirin	623 (24.5%)	209 (24.6%)	208 (22.5%)	−2.0 (−5.1 to 1.1)
P2Y12 inhibitor	126 (6.1%)	42 (6.5%)	51 (5.5%)	0.5 (−1.7 to 2.6)
Warfarin	185 (7.3%)	75 (8.8%)	56 (6.0%)	−1.2 (−3.1 to 0.1)
Direct Thrombin Inhibitor	16 (0.6%)	5 (0.6%)	12 (1.3%)	0.7 (−0.1 to 1.5)
Non-Steroidal Anti-Inflammatory Drugs	131 (5.5%)	50 (6.1%)	44 (4.8%)	−0.3 (−2.1 to 1.5)
Proton Pump Inhibitor	656 (26.2%)	229 (27.2%)	220 (23.8%)	−2.4 (−5.6 to 0.9)
Histamine-2 Receptor Antagonist	121 (4.9%)	40 (4.7%)	55 (5.9%)	1.0 (−0.7 to 2.8)
Direct Oral Anticoagulant	58 (2.3%)	15 (1.8%)	41 (4.4%)	2.2 (0.7 to 3.6)

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Table 2:

Input variables in the final dataset. Structured datafields mapped to specific categories for past medical history using Clinical Classifications Software Refined (CCS), and medication classes specified by Epic.

Variable class	Variables
Demographics (2)	Age at encounter Sex
Vital Signs (8)	Diastolic blood pressure Systolic blood pressure Pulse Pulse oximetry Mean arterial pressure Respiratory Rate Shock Index Temperature
Nursing Assessment (3)	Glasgow Coma Score Eye Component Glasgow Coma Score Motor Component Glasgow Coma Score Verbal Component
Lab Values (20)	Albumin Alkaline Phosphatase Alanine Aminotransferase Aspartate Aminotransferase Total Bilirubin Blood urea nitrogen Chloride Bicarbonate Creatinine Hematocrit Hemoglobin International normalized ratio Potassium Mean corpuscular hemoglobin Mean corpuscular hemoglobin concentration Mean corpuscular volume Platelet count Red cell distribution width Sodium White blood cell count Blood Urea Nitrogen / Creatinine
Phecodes (48)	Cancer of Liver and Intrahepatic Bile Duct Pancreatic Cancer Malignant Neoplasm (Other) Neurofibromatosis Myeloproliferative Disease Multiple Myeloma Type 2 Diabetes Adult failure to thrive Other anemia Disease of spleen Specific nonpsychotic mental disorders due to brain damage Memory loss Schizophrenia Psychosis Suicidal ideation Anxiety disorder Cognitive impairment Convulsions Disorders of vitreous body Nonrheumatic aortic valve disorders Coronary atherosclerosis Second degree AV block Paroxysmal supraventricular tachycardia Congestive heart failure Transient cerebral ischemia Noninfectious disorders of lymphatic channels Varicose veins Emphysema Chronic bronchitis

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Variable class	Variables
	Gastroparesis Regional enteritis Noninfectious gastroenteritis Cirrhosis of liver without mention of alcohol Primary biliary cirrhosis Ascites (non-malignant) Jaundice Cholelithiasis Hemorrhage of gastrointestinal tract Renal sclerosis NOS Hematuria Other disorders of prostate Rash and other nonspecific skin eruption Pruritis and related conditions Cardiac and circulatory congenital anomalies Congenital osteodystrophies Muscle weakness Symptoms involving nervous and musculoskeletal systems Other abnormal blood chemistry
Medication Classes (25)	Alpha adrenergic blocking agents Alpha beta adrenergic blocking agents Antacids Antihistamines SGLT2 inhibitors PPARγ agonist Anti-Neoplastic Systemic Enzyme Inhibitors Antithyroid Preparations Antivirals Beta adrenergic blocking agents Digitalis Glycosides Heparin and Related Preparations Iron replacement Loop diuretics Norepinephrine and Dopamine Reuptake Inhibitors (NDRI) Vasoconstrictors (Nasal) Opioid Antagonists Beta 3 Adrenergic Receptor Agonist Potassium Sparing Diuretics Quinolone Antibiotics Sodium saline Preparation Vitamin B1 preparation Vitamin B12 Preparation Zinc Replacement Aspirin

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Table 3:

Components of the Glasgow Blatchford Score and Oakland Score in the external validation dataset. Details about these scores in Appendix A5.

External Validation (N=926)	Glasgow-Blatchford Score	Oakland Score
	Mean (SD)	
Total Score	6.3 (4.6)	16.8 (0.24)
	Proportion (%)	
Pulse	0: 74.3% 1: 25.7%	0: 42.1% 1: 28.7% 2: 14.8% 3: 14.4%
Systolic Blood Pressure	0: 84.4% 1: 8.1% 2: 4.4% 3: 3.0%	0: 14.8% 2: 41.9% 3: 16.2% 4: 24.1% 5: 3.0%
Hemoglobin	0: 35.6% 1: 16.0% 3: 9.6% 6: 38.8%	0: 4.2% 4: 25.4% 8: 22.7% 13: 18.0% 17: 16.7% 22: 13.0%
Blood Urea Nitrogen	0: 43.4% 2: 13.4% 3: 11.2% 4: 28.5% 6: 3.4%	Not Included in Oakland Score
Age	Not Included in Glasgow-Blatchford Score	0: 9.1% 1: 44.8% 2: 46.1%
Gender	Female: 44.2% Male: 55.8%	Female: 44.2% Male: 55.8%
Previous LGIB	Not Included in Glasgow-Blatchford Score	9.2%
Melena	30.8%	Not Included in Oakland Score
Liver Failure	14.4%	Not Included in Oakland Score
Cardiac Failure	15.7%	Not Included in Oakland Score
Syncope	6.6%	Not Included in Oakland Score

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Table 4:

Area under the receiver operating characteristic curve (AUC) for the gradient boosted tree machine learning model vs. deep learning model vs. Glasgow-Blatchford Score and Oakland Score.

Composite Endpoint[a]	Internal Validation AUC with 99% Confidence Interval	External Validation AUC with 99% Confidence Interval	p-value
XGBoost Machine Learning Model	0.91 (0.88–0.93)	0.92 (0.90–0.95)	Ref
Neural Network Model	0.90 (0.88–0.93)	0.92 (0.90–0.94)	0.28
Glasgow-Blatchford score		0.89 (0.86–0.92)	<0.001
Oakland Score		0.89 (0.86–0.92)	<0.001

[a]Composite Outcome: Red-blood-cell transfusion, hemostatic intervention (endoscopic, interventional radiologic, or surgical), or 30-day mortality

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Table 5:

Comparison of Specificity and Number of Correctly Identified Very-Low-Risk patients with Machine Learning Model versus Deep Learning Model and Clinical Risk Scores at very-low-risk threshold of 99% sensitivity.

Model	Specificity at 99% Sensitivity	p-value	Number of Very-Low-Risk Patients Identified in External Validation (N=926)
XGBoost Machine Learning Model	37.9%	Ref	211
Neural Network Model	28.2%	0.002	157
Glasgow-Blatchford score=0	19.6%	<0.001	103
Oakland Score=8	11.7%	<0.001	65